

LESSON

4

The Active Loop

Check, repeat, and respond

MISSION Can you trace three loop cycles and stop before a fourth one begins?

Learning targets

CHECK	BY THE END OF THIS LESSON, I CAN...
[]	Explain opModelsActive().
[]	Compare if and while checks.
[]	Identify the loop body.
[]	Explain why fresh input keeps TeleOp responsive.

Words to know

VOCABULARY boolean | true | false | if | while | condition | iteration | loop body | braces

Code focus

```
if (opModeIsActive()) {
    while (opModeIsActive()) {
        // READ
        // COMMAND
        // REPORT
    }
}
```

Before the video: commit to a prediction

MY PREDICTION If opModelsActive() becomes false at the top of a cycle, which loop statements run next?

Confidence before the lesson: 1 2 3 4 5 (circle one)

The POWERING ON safety contract

NON-NEGOTIABLE The sequence is completed before every live-motor test. If a step is skipped, remain disabled and restart at Step 1.

STEP	ACTION	TEAM CHECK
1	SECURE THE ROBOT Use stable blocks. Keep driven wheels and moving mechanisms clear of the table and floor.	[]
2	CONFIRM DISABLED Select the correct robot and OpMode, but do not press PLAY.	[]
3	LOOK AROUND Remove hands, tools, wires, scraps, and loose objects from moving areas.	[]
4	MAKE SURE ALL ARE CLEAR Everyone steps outside the boundary; secure hair, clothing, lanyards, strings, and jewelry.	[]
5	CALL AND WAIT Call POWERING ON! loudly. Pause long enough for everyone to stop, look up, and step away.	[]
6	VISUALLY CONFIRM, THEN ENABLE Scan the full area again. Press PLAY only when everyone and everything is clear.	[]

Callout script

BEFORE PLAY Robot is secured and DISABLED. Look around. Everyone clear? POWERING ON! ... Wait ... Area is visibly clear. Enabling now.

BEFORE CONTACT STOP pressed. Driver Station shows DISABLED. Hands may approach only after the operator confirms disabled.

Student agreement

I understand that anyone may call STOP and the operator will disable immediately. I will not approach or touch the robot until DISABLED is visually confirmed.

Student signature: _____ Date: _____

Reference program: BasicTwoMotorTeleOp.java

Use this numbered reference throughout Lessons 1-6. The comments divide the program into startup and repeating control sections.

```

01 package org.firstinspires.ftc.teamcode;
02
03 import com.qualcomm.robotcore.eventloop.opmode.LinearOpMode;
04 import com.qualcomm.robotcore.eventloop.opmode.TeleOp;
05 import com.qualcomm.robotcore.hardware.DcMotor;
06
07 @TeleOp(name="Basic Two Motor TeleOp", group="Iterative Opmode")
08 public class BasicTwoMotorTeleOp extends LinearOpMode {
09     private DcMotor leftMotor;
10     private DcMotor rightMotor;
11
12     @Override
13     public void runOpMode() {
14         leftMotor = hardwareMap.get(DcMotor.class, "left_motor");
15         rightMotor = hardwareMap.get(DcMotor.class, "right_motor");
16
17         // Set one motor REVERSE if the physical drivetrain requires it.
18         // rightMotor.setDirection(DcMotor.Direction.REVERSE);
19
20         telemetry.addData("Status", "Initialized");
21         telemetry.update();
22         waitForStart();
23
24         if (opModeIsActive()) {
25             while (opModeIsActive()) {
26                 double leftPower = -gamepad1.left_stick_y;
27                 double rightPower = -gamepad1.right_stick_y;
28                 leftMotor.setPower(leftPower);
29                 rightMotor.setPower(rightPower);
30                 telemetry.addData("Left Power", leftPower);
31                 telemetry.addData("Right Power", rightPower);
32                 telemetry.update();
33             }
34         }
35     }
36 }

```

CODE DETECTIVE Circle `waitForStart()`. Box the while condition. Underline both `setPower()` calls. Draw a star beside each telemetry line.

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Guided Video Notes

The Active Loop

DIRECTIONS At each timestamp, pause, think silently, discuss, and then record evidence from the code or whiteboard visual.

PAUSE	QUESTION	MY CODE / VISUAL EVIDENCE
0:17	What values can <code>opModelsActive()</code> return?	
0:26	How often does the outer if check?	
0:36	How often does the while condition check?	
0:45	What do the braces identify?	
1:05	What three jobs happen in each cycle?	
1:44	Why read the joysticks again every cycle?	
1:53	What causes the loop to exit?	

MOST IMPORTANT IDEA The idea I would teach to a teammate is:

LESSON**4****Hands-On Team Activity***The Active Loop***Materials and setup**

READY	ITEM
[]	Printed BasicTwoMotorTeleOp.java with space to mark braces.
[]	Role cards: CHECK, READ, COMMAND, REPORT, DRIVER, CODE TRACER, and STOP.
[]	Program token; TRUE/FALSE cards; three left/right joystick-value cards.
[]	Optional secured robot and telemetry display for the final extension.
[]	POWERING ON safety poster if the live extension will be used.

Team roles

ROLE	JOB
CHECK	Reads opModelsActive() and chooses repeat or exit.
READ	Accepts the newest joystick-value card.
COMMAND	Sends the current values to the motors.
REPORT	Announces the telemetry values.
Driver	Changes the input card between cycles.
Code tracer	Points to each represented code line.
STOP	Receives the token when the condition is false.
Recorder	Documents each completed iteration.

Activity sequence

STEP	ACTION
1	Ask students to box the condition and connect the opening and closing braces of the while statement.
2	Point to the outer if and inner while. Ask how many times each condition is evaluated when execution reaches it.
3	Give the driver a different input pair before each cycle. Pass the token CHECK -> READ -> COMMAND -> REPORT -> CHECK.
4	Replace TRUE with FALSE when the token returns to CHECK. Ask what happens before another READ.
5	If using the robot, secure it and run the complete POWERING ON sequence before PLAY. Keep sticks centered and observe telemetry changing with a gentle input.

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Lab Record and Exit Ticket

The Active Loop

Student/team: _____ Date: _____ Robot: _____

BEFORE PLAY	AFTER TEST
☐ Robot secure ☐ DISABLED ☐ Area clear ☐ POWERING ON called ☐ Wait completed ☐ Visual clear	☐ STOP pressed ☐ DISABLED visually confirmed Safety lead initials: _____

Control-loop cycle record

CYCLE	ACTIVE?	NEW INPUT L / R	COMMAND / REPORT / NEXT
1	TRUE	+0.4 / +0.4	Motors: ____ Telemetry: ____ Next: ____
2	TRUE	+0.6 / +0.2	Motors: ____ Telemetry: ____ Next: ____
3	TRUE	0.0 / 0.0	Motors: ____ Telemetry: ____ Next: ____
4	FALSE	New card withheld	Does loop body run? ____ Next: ____
Student-created	____	____ / ____	Prediction: _____

Debrief

#	QUESTION	MY RESPONSE
1	Which jobs repeat?	
2	Why is an active check placed before the body?	
3	What would happen if inputs were read only once?	
4	Does motor power 0.0 end the loop?	

Exit ticket

ANSWER BEFORE LEAVING Describe one full loop cycle and explain exactly what prevents the next cycle after STOP.
